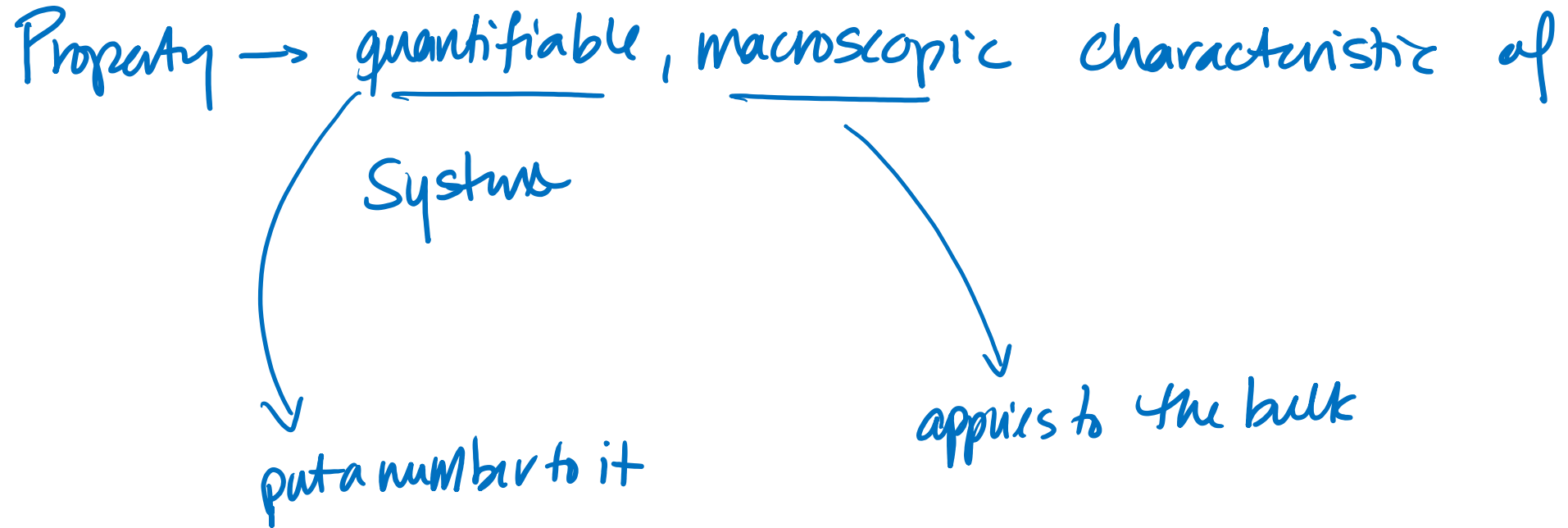


ME 300: Engineering Thermodynamics

Properties and States



What is a property?





What are some
examples of
properties?

Property examples and thermodynamic properties

Mass

Volume

Temperature

Pressure

humidity

color

internal energy

entropy

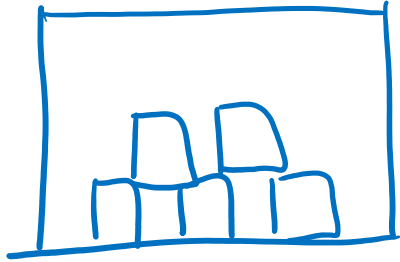
heat capacity

↳ have to do with
the energy of the
system.

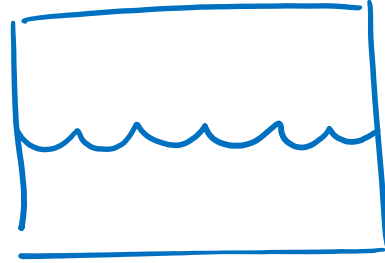
Thermodynamic state

State of a system is defined by all the system properties.

Example: ice in Tupperware



State 1



State 2



What properties
changed as the ice
melted?

Example: ice in Tupperware

State 1

Temp

Volume

State 2

T ↑

∇

Phase - solid → liquid

$$M_1 = M_2$$